



Comparative analysis of production traits and culling outcomes in adult Holstein and Montbéliarde × Holstein crossbred cows

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ABSTRACT

Holstein (H) cows and first-generation Montbéliarde × Holstein (MH) crossbred cows were compared for lactation performance, culling reasons, and cull price under the same intensive dairy production system. Data were obtained from a single commercial dairy farm in Jiangsu Province, China, including lactation records from 504 H cows and 367 MH cows and culling records from 345 H cows and 178 MH cows born between 2013 and 2022. Lactation traits from the first 3 parities were analyzed using linear mixed models, and culling reasons were evaluated using chi-squared tests and multinomial logistic regression, with low milk yield as the reference outcome. Overall milk yield and milk component yields did not differ between breeds across lactations (differences <1.6%). However, H cows exhibited a higher peak milk yield (48.3 vs. 46.9 kg), whereas days to peak milk yield were similar between breeds, with only a small numerical difference. The distribution of culling reasons differed markedly between breeds, with body condition abnormalities accounting for a larger proportion of culling in MH cows (41.6% vs. 15.4%) and a higher proportion of culling events due to death in H cows (20.6% vs. 10.7%), and overall removal risk was higher in H cows than in MH cows when expressed per 100 cow-yr at risk. Multinomial regression showed that calving season, parity, and age (modeled within parity) were significant predictors of multiple culling outcomes, whereas inclusion of BW in sensitivity analyses did not materially alter breed-related effects. Cull price was positively associated with BW but did not differ between breeds. These results indicate that although H and MH cows achieve comparable cumulative milk production, breed differences are primarily reflected in lactation curve characteristics and culling structure rather than total productivity.

Key words: Montbéliarde × Holstein, crossbreeding, lactation performance, culling reasons

INTRODUCTION

Over the past several decades, continuous genetic improvement and advances in dairy management have markedly increased milk production in modern Holstein (H) cows (Oltenuacu and Broom, 2010; Brito et al., 2021). However, the metabolic burden associated with high yield has also predisposed cows to a higher incidence of mastitis, lameness, reproductive disorders, and other health problems, ultimately accelerating culling and replacement rates (De Vries, 2017). These health challenges not only compromise production efficiency but also elevate involuntary culling, thereby reducing the overall economic performance of dairy farms (Halasa et al., 2007; Adamie et al., 2023). In the context of increasing volatility in milk prices, the relatively stable value of cull cows and calves has attracted growing attention, making the integration of dairy and beef production a more appealing strategy (Domingo et al., 2015; Berry, 2021). Consequently, the slaughter performance and economic value of culled cows have become important components of dairy production systems (Heins et al., 2012a; Domingo et al., 2015; Moreira et al., 2021).

To balance milk yield, health, and long-term sustainability, many countries have adopted crossbreeding strategies aimed at improving fertility (Heins et al., 2006a, 2012b; Knob et al., 2016), milk composition (Freyer et al., 2008; Palladino et al., 2010; Solarczyk et al., 2024), disease resistance (Mendonça et al., 2014; Zhang et al., 2024; Macedo et al., 2026), and survival (Sørensen et al., 2008; Cartwright et al., 2011) in dairy cattle. Extensive crossbreeding programs involving breeds such as Montbéliarde, Norwegian Red, Scandinavian Red, Simmental, and Jersey have been implemented across Europe (Freyer et al., 2008; Sørensen et al., 2008; Kargo et al., 2021), North America (Heins and Hansen, 2012), and other regions (Pipino et al., 2025). These studies consistently

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

demonstrate that well-designed crossbreeding systems can maintain competitive milk performance while enhancing reproductive efficiency, body condition, health traits, and overall herd longevity (Heins et al., 2006a, 2012a; Dezetter et al., 2017). Such advantages contribute to lower involuntary culling rates and improved economic returns (Heins et al., 2012a,b; Shonka-Martin et al., 2019). As a dual-purpose breed with strong functional traits and favorable reproductive performance (Heins and Hansen, 2012; Mendonça et al., 2014; Shonka-Martin et al., 2019), Montbéliarde cattle have shown considerable potential in international crossbreeding systems and have become a focal point of recent research.

Nevertheless, the outcomes of crossbreeding are often region-specific and influenced by climate (El-Tarabany et al., 2018), feeding strategies (Johansen et al., 2024), nutritional management (Rahman et al., 2024), and local disease ecology (Quénou et al., 2023). To our knowledge, research on dairy cattle crossbreeding in China remains limited. More importantly, there is a lack of studies conducted under unified conditions (within the same farm and management system and using animals with comparable birth and rearing backgrounds) to systematically compare purebred and crossbred cows. Even fewer studies have simultaneously evaluated lactation performance, culling reasons, and cull-cow economic value within the same production environment.

Therefore, the objective of this study was to compare (1) lactation performance, (2) culling reasons, and (3) cull price between pure H cows and first-generation Montbéliarde × Holstein (MH) crossbred cows raised under the same production system. The results are expected to provide guidance for optimizing crossbreeding strategies in Chinese dairy farms and to support improvements in herd health, production efficiency, and sustainability of the dairy industry.

MATERIALS AND METHODS

Data Collection

A total of 504 purebred H dairy cows and 367 first-generation crossbreds (Montbéliarde sire × Holstein dam), all born in Xinyi City, Jiangsu Province, China, between November 2015 and June 2022, were used for lactation performance analysis. The cows had ad libitum access to feed and water. Heifers and multiparous cows were housed separately. Lactating cows were fed a TMR consisting mainly of corn silage, oat hay, alfalfa hay, beet pulp, corn and soybean meal, and cottonseed, with 3 feedings per day. Although the composition of the TMR varied slightly across years and seasons, the overall composition remained consistent. The cows were milked 3 times a day at ~0530, 1350, and 1930 h, using a parallel milking parlor.

Production performance data for first, second, and third lactations, including test-day milk yield records collected monthly, as well as culling data, were retrieved from the farm's DHI analysis center database. Culling records were collected during 2023 and 2024 from the study herd. All culled cows included in the analysis were born between 2013 and 2022 and included 345 H cows and 178 first-generation crossbreds. The main culling reasons were body condition abnormalities, feet or leg problems, death, low production, reproductive problems, mastitis, udder conformation issues, diarrhea, abortion or premature calving, various diseases, and injuries. Body condition abnormalities was a farm-specific culling category used to describe cows removed due to persistently poor body condition that could not be corrected through routine management interventions. This classification was not based on a standardized body condition scoring system and may overlap with underlying health disorders when no single clinical diagnosis was identified as the primary cause of culling. For the purpose of this study, the culling reasons were categorized into body condition abnormalities, feet or leg problems, death, low production, reproductive problems, mastitis, udder conformation issues, and other (which includes diarrhea, abortion, injury, and other diseases that met the culling criteria). Low milk yield was used as the reference category in all multinomial models. When performing multinomial logistic regression analysis, mastitis and udder conformation issues were merged into the "other" category. Although a cow might have been culled due to multiple causes, the primary culling reason was considered for analysis. The culling weight was recorded at livestock auction markets, and the unit price of culling weight was calculated by dividing the total price by the weight.

Statistical Analysis

Lactation performance traits were analyzed at the lactation level, with one record per cow per parity. For each lactation, 305-d milk yield, 305-d mature equivalent milk yield, actual lactation milk yield, total protein yield, and total fat yield were obtained directly from the farm's DHI recording system, where these traits are routinely calculated based on monthly test-day records. Peak milk yield and days to peak yield were identified from observed test-day records as the maximum recorded daily milk yield and its corresponding day in milk.

To further characterize lactation dynamics, lactation curves were modeled using the Wood incomplete gamma function (Wood, 1967) based on monthly test-day milk yield records. Observations with missing cow identification, missing lactation number, DIM <5 or >305, or nonpositive milk yield were excluded. Each lactation was required to include at least 5 test-day records and

extend beyond 250 DIM to ensure reliable curve fitting. The Wood model was fitted separately for each lactation using nonlinear regression in SAS (version 9.4; SAS Institute, Cary, NC). Parameters were constrained to positive values, and only successfully converged curves were retained. The 3 parameters describe initial yield level (**a**), ascending slope to peak (**b**), and postpeak decline rate (**c**). Days to peak, peak milk yield, and 305-d milk yield were derived from the fitted curves. Biologically implausible estimates were excluded (peak day <20 or >300 d, peak yield <5 or >80 kg/d, or 305-d milk yield <3,000 or >20,000 kg).

Derived lactation traits and Wood parameters were analyzed using linear mixed models (PROC MIXED, REML). Breed, parity group (1 vs. 2–3), and calving season were included as fixed effects, and cow was fitted as a random intercept to account for repeated lactations within cow. Degrees of freedom were estimated using the Kenward–Roger method. Least squares means were compared, and statistical significance was declared at $P < 0.05$.

The study population included all H and MH cows present in the herd between January 1, 2023, and December 31, 2024. In total, 699 H cows and 335 MH cows contributed time at risk during the study period. To define time at risk, cows contributed follow-up time from the later of herd entry date or January 1, 2023, until the earliest of culling, death, sale, or December 31, 2024. Removal events were considered only if they occurred within the observation window. Cows that remained in the herd at the end of follow-up were treated as right-censored. Accordingly, the risk set consisted of all cows contributing follow-up time during the study period, regardless of whether a removal event occurred. Cow-years at risk were calculated as the sum of individual follow-up time divided by 365.25. Consequently, cause-specific incidence rates were expressed per 100 cow-yr at risk, thereby accounting for both the number of cows at risk and duration of follow-up. Breed differences in cause-specific incidence rates were assessed using Poisson regression models with a log(cow-years) offset to account for differences in time at risk. In addition, false discovery rate was controlled using the Benjamini–Hochberg procedure. Overall removal risk between breeds was further evaluated using Cox proportional hazards regression.

For the analysis of culling reasons, chi-squared tests were first conducted for descriptive comparisons and overall association screening, followed by multinomial logistic regression to provide adjusted quantitative interpretations. During multinomial logistic regression, the number of observations for mastitis and udder conformation issues in the crossbred group was too small, which resulted in model nonconvergence and instability;

therefore, these categories were combined into “other.” Accordingly, the final regression included 6 culling categories: body condition abnormalities, feet or leg problems, death, low production, reproductive problems, and other. To limit collinearity between age and parity, age at culling was modeled as a parity-specific variable nested within parity group. Specifically, age at first calving was used for first-parity cows, age at second calving for second-parity cows, age at third calving for cows in parity groups 3 to 4, and age at fifth calving for cows in parity group ≥ 5 . This approach allowed direct evaluation of age-related effects within parity strata while reducing redundancy between chronological age and reproductive history. Parity was grouped as 1, 2, 3 to 4, and >5 , lactation stage was grouped as early (1–100), mid (101–200), and late (>200). Because BW was not recorded for cows that died on-farm (no market weight available), the primary model included all animals and did not force imputation for missing BW. A secondary model excluded the 90 dead cows and additionally included BW to assess the robustness of breed effects and other covariate estimates.

Cull price (Chinese yuan [CNY]/kg) was analyzed using a linear mixed model. Breed (Holstein vs. Montbéliarde $\times$ Holstein) and cull year (2023 vs. 2024) were included as fixed class effects, whereas age at culling (mo), DIM, and BW (kg) were included as continuous covariates. Sales market was included as a random intercept to account for clustering of cull prices within markets. To evaluate the robustness of breed effects, a sensitivity analysis was conducted by fitting an additional model excluding BW. Cull price was recorded and analyzed in CNY. For the convenience of international comparison, average exchange rates in 2024 were provided: US\$1 = 7.12 CNY and €1 = 7.72 CNY.

RESULTS

Production Traits and Lactation Curve Characteristics

The lactation performance of H cows and MH crossbred cows is summarized in Table 1. Overall, no significant differences were observed between the 2 groups for most milk production traits. The 305-d milk yield was comparable between H and MH cows, averaging 10,849.1 and 10,890.1 kg, respectively. Similarly, 305-d mature equivalent milk yield did not differ between breeds, with MH cows producing numerically higher yields (12,074.0 kg) than H cows (11,884.7 kg), although the difference was not significant. Actual lactation milk yield also did not vary between groups, averaging 15,044.5 kg in H cows and 14,766.1 kg in MH cows. Milk component yields followed the same pattern, with no significant differences in total protein yield (484.6 kg

Table 1. Least squares means ($\pm$ SEM) of lactation performance traits for Holstein (H) and Montbéliarde $\times$ Holstein (MH) cows¹

Trait	H	SEM	MH	SEM
305-d milk yield (kg)	10,849.1	486	10,890.1	496
305-d mature equivalent milk yield (kg)	11,884.7	522	12,074	532
Actual lactation milk yield (kg)	15,044.5	861	14,766.1	895
Total protein yield (kg)	484.6	29.9	483.7	31.1
Total fat yield (kg)	521.1	29.4	519.9	30.6
Peak milk yield (kg)	48.3 ^a	1.6	46.9 ^b	1.6
Days to peak yield (d)	92.0	11.1	93.5	11.3

^{a,b}Values within a row with different superscripts differ ($P < 0.05$).

¹n = 504 for Holstein; n = 367 for Montbéliarde $\times$ Holstein.

for H vs. 483.7 kg for MH) or total fat yield (521.1 kg for H vs. 519.9 kg for MH).

In contrast, significant breed effects were detected for peak lactation traits ($P < 0.05$). H cows exhibited a higher peak milk yield (48.3 kg) compared with MH cows (46.9 kg). Days to peak milk yield did not differ significantly between breeds, with only a small numerical difference (92.0 vs. 93.5 d) that was not significant.

Together, these results indicate that while overall lactation performance was similar between H cows and MH cows, crossbred cows showed a lower peak milk yield relative to pure H cows.

Adjusted lactation curves by parity are presented in Figure 1, and Wood model parameters and derived traits are shown in Supplemental Table S1, see Notes. In first lactation, H cows had a higher initial yield parameter (a) than MH cows (16.33 vs. 14.90; $P < 0.001$), resulting in greater peak milk yield (40.6 vs. 38.6 kg/d; $P < 0.001$) and higher 305-d milk yield (11,167 vs. 10,601 kg; $P < 0.001$). Parameters b and c, as well as days to peak (121.3 vs. 118.8 d), did not differ between breeds. In second and third lactations, no significant breed differences were detected for Wood parameters, peak yield (48.2 vs. 48.6 kg/d), days to peak (82.2 vs. 77.4 d), or 305-d milk yield (12,011 vs. 12,255 kg). Across breeds, multiparous cows reached higher peak yields and peaked earlier than primiparous cows.

Season-specific adjusted curves are shown in Figure 2 and corresponding estimates in Supplemental Table S2, see Notes. In spring, H cows had higher peak milk yield than MH cows (46.1 vs. 43.7 kg/d; $P = 0.039$), whereas days to peak and 305-d milk yield were similar. In summer, MH cows exhibited a higher peak milk yield than H cows (43.2 vs. 39.7 kg/d; $P = 0.0302$). However, days to peak (106.3 vs. 116.4 d) and 305-d milk yield (11,192 vs. 10,489 kg) did not differ significantly between breeds. Wood parameters (a, b, and c) were also similar between groups. In autumn, no breed differences were observed for peak yield, days to peak, or cumulative production. In winter, H cows produced more 305-d milk than MH cows (12,107 vs. 11,679 kg; $P = 0.0437$), whereas peak yield

and days to peak did not differ. Across seasons, summer calving was associated with delayed peak and lower production, particularly in H cows, whereas winter calving resulted in the highest 305-d milk yield in both breeds.

Breed differences in lactation dynamics were evident primarily in first lactation and in specific calving seasons. However, overall lactation curves were broadly similar between H and MH cows across parities and seasons.

Culling Incidence Rates

Incidence rates of primary culling reasons expressed per 100 cow-yr at risk are presented in Table 2. Overall removal incidence was higher in H cows than in MH cows (52.9 vs. 37.4 removals per 100 cow-yr; $q < 0.001$). Consistent with this pattern, Cox proportional hazards analysis indicated a significantly higher overall culling hazard for Holstein cows compared with MH cows (hazard ratio = 1.55, 95% CI: 1.29–1.86; $P < 0.001$).

Marked breed differences were observed in the incidence of specific culling reasons. The MH cows showed a substantially higher incidence of culling due to body condition abnormalities compared with Holstein cows (14.9 vs. 7.8 per 100 cow-yr; $q = 0.002$). In contrast, Holstein cows exhibited higher incidence rates of death (10.9 vs. 4.0 per 100 cow-yr; $q = 0.001$) and reproductive problems (5.5 vs. 2.3 per 100 cow-yr; $q = 0.030$). Incidence rates for feet or leg problems, low production, mastitis, and udder conformation issues tended to be numerically higher in Holstein cows, although these differences did not reach statistical significance after false discovery rate adjustment ($q \geq 0.056$).

Chi-Square Test of Culling Reasons

Whereas Table 2 presents population-based incidence rates, Table 3 describes the relative distribution of primary removal reasons among culled cows. The distribution of primary reasons for culling differed markedly between H cows and MH cows (Table 3). These proportions reflect the distribution of causes among culled cows within each

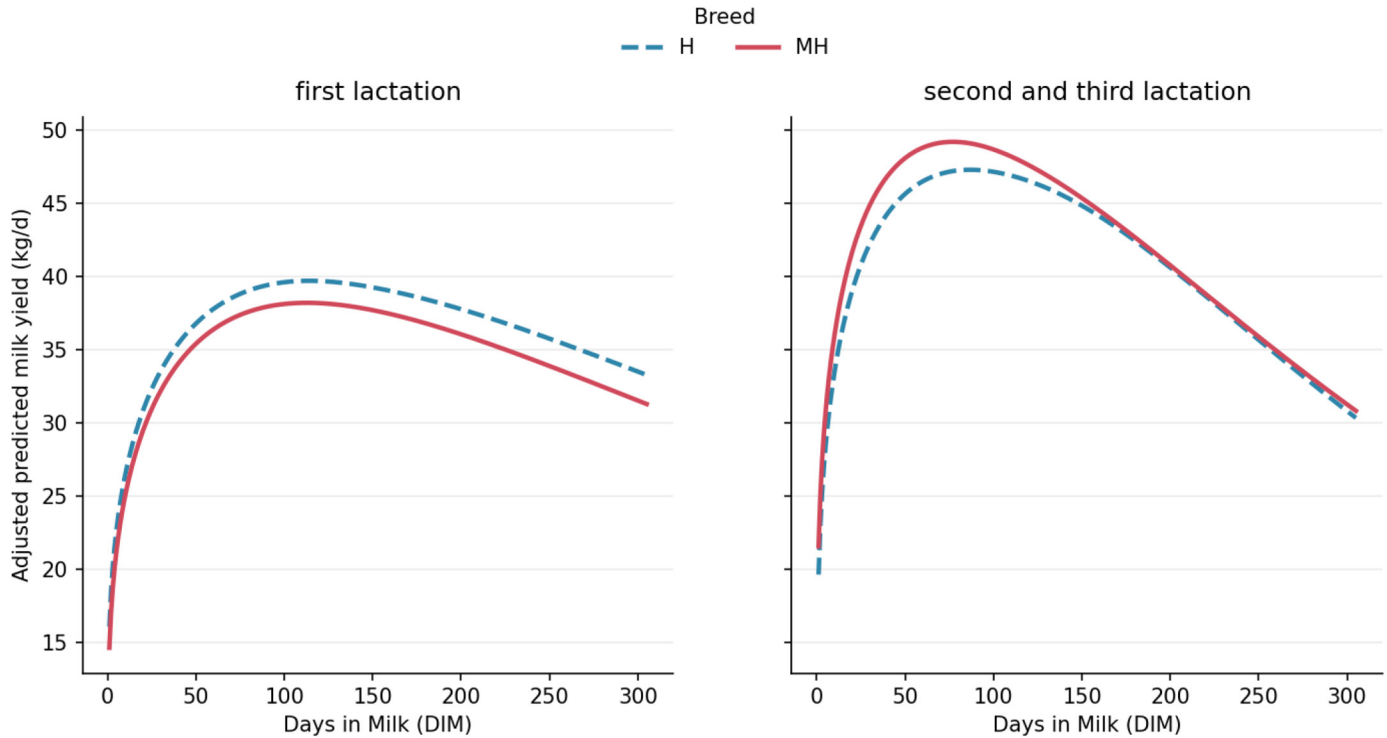


Figure 1. Adjusted lactation curves derived from the Wood model for Holstein (H) and Montbéliarde × Holstein (MH) cows by parity group (first lactation vs. second and third lactations). Predicted milk yield (kg/d) was obtained from linear mixed models using LSM.

breed and do not represent population-based culling risk. A total of 523 cows were culled during the observation period, including 345 H cows and 178 MH cows. Overall, the composition of culling reasons differed significantly

between breeds ($\chi^2 = 46.49$; $P < 0.001$). Significant breed differences ($P < 0.001$) were detected for body condition abnormalities and death loss, whereas other culling categories showed no differences between groups.

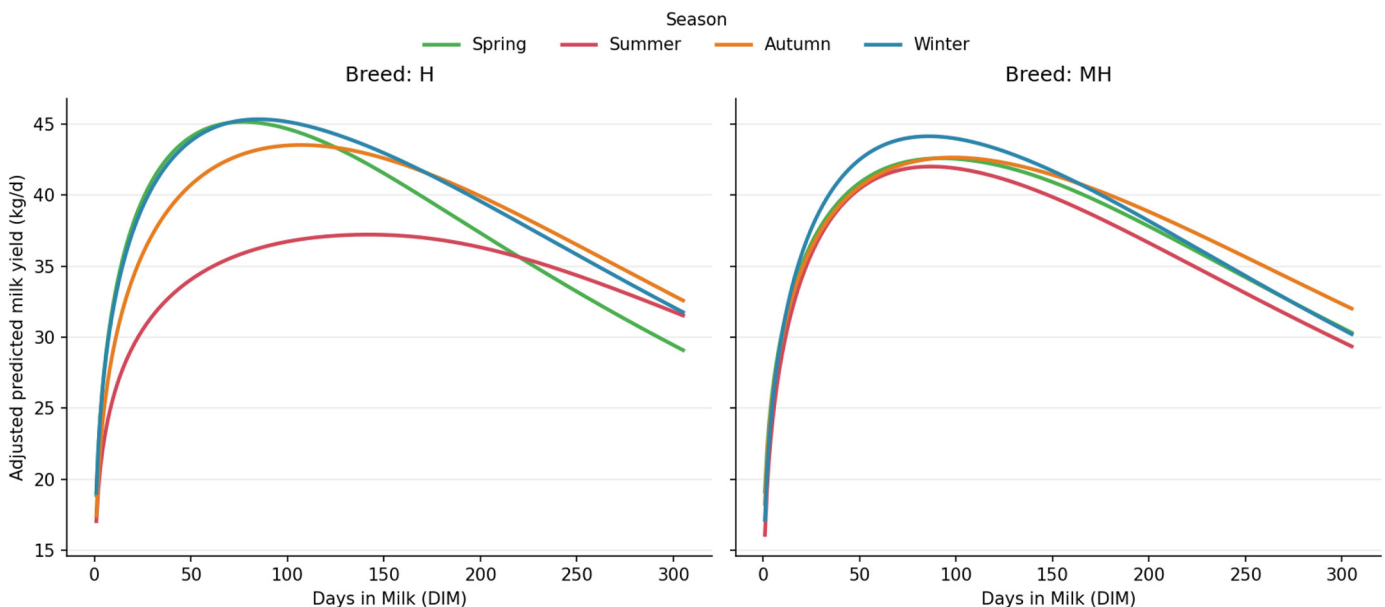


Figure 2. Adjusted lactation curves derived from the Wood model for Holstein (H) and Montbéliarde × Holstein (MH) cows by calving season (spring, summer, autumn, and winter). Predicted milk yield (kg/d) represents LSM from linear mixed models.

Table 2. Incidence of primary culling reasons in Holstein (H) and Montbéliarde × Holstein (MH) cows, expressed per 100 cow-yr at risk

Primary reason for culling	H		MH		q-value ²
	N	Incidence rate ¹	N	Incidence rate	
Body condition abnormalities	51	7.8 ^b	71	14.9 ^a	0.002
Feet or leg problems	65	10.0	33	6.9	0.087
Low production	56	8.6	25	5.2	0.056
Mastitis	23	3.5	7	1.5	0.056
Other	21	3.2 ^a	5	1.1 ^b	0.048
Reproductive problems	36	5.5 ^a	11	2.3 ^b	0.030
Udder conformation issues	22	3.4	7	1.5	0.063
Death	71	10.9 ^a	19	4.0 ^b	0.001
Total ³	345	52.9	178	37.4	<0.001

^{a,b}Within each row, different superscripts (a, b) indicate a significant difference between breeds ($q < 0.05$).

¹Values represent incidence rates per 100 cow-yr at risk.

²Breed differences for each culling reason were evaluated using multivariable Poisson regression with a log(cow-yr) offset. Benjamini-Hochberg false discovery rate-adjusted q-values are shown.

³Overall removal risk was further assessed by Cox regression, indicating higher hazard in H than MH (hazard ratio = 1.55, 95% CI: 1.291–1.860; $P < 0.001$).

Body condition abnormalities represented a substantially larger proportion of culling events in MH cows (39.9%) compared with H cows (14.8%; $P < 0.001$). In contrast, the proportion of individuals culled due to death was significantly higher in H cows (20.6%) than in MH cows (10.7%; $P < 0.001$).

Overall, these results indicate that although several culling categories occurred at similar incidence rates between breeds (Table 2), the composition of removal reasons differed between H and MH cows when considering only culled animals (Table 3).

Multinomial Logistic Regression Analysis of Culling Reasons (Excluding BW)

Because BW was not recorded for cows that died on-farm (no market weight available), the primary model included all animals and did not force imputation for

missing BW. Multinomial logistic regression identified several significant predictors associated with culling reasons when low milk yield was used as the reference category. Adjusted associations between predictors and culling reasons are summarized in Table 4, with full model outputs provided in Supplemental Table S3, see Notes.

The MH crossbred cows had higher odds of being culled due to body condition abnormalities than H cows (odds ratio [OR] = 2.83, 95% CI: 1.39–5.78; $P = 0.004$). Early lactation was associated with a markedly lower likelihood of culling for body condition abnormalities compared with late lactation (OR = 0.19; $P < 0.001$).

For feet or leg problems, early lactation was associated with reduced odds of culling relative to late lactation (OR = 0.37; $P = 0.007$), whereas breed effects were not significant in the primary model.

For death as a culling outcome, spring calving showed a near-significant increase in risk compared with winter

Table 3. Primary culling reasons in Holstein (H) and Montbéliarde × Holstein (MH) cows (% within breed)^{1,2}

Primary reason for culling	H		MH		Total	
	N	%	N	%	N	%
Body condition abnormalities	51	14.8 ^b	71	39.9 ^a	122	23.3
Feet or leg problems	65	18.8	33	18.5	98	18.7
Death	71	20.6 ^a	19	10.7 ^b	90	17.2
Low production	56	16.2	25	14.0	81	15.5
Reproductive problems	36	10.4	11	6.2	47	9.0
Mastitis	23	6.7	7	3.9	30	5.7
Udder conformation issues	22	6.4	7	3.9	29	5.5
Other	21	6.1	5	2.8	26	5.0
Total	345	100.0	178	100	523	100.0

^{a,b}Percentages within a row with different superscripts differ significantly between breeds ($P < 0.05$).

¹Holstein, $n = 345$; Montbéliarde × Holstein, $n = 178$; Total, $n = 523$.

²Overall $\chi^2 = 46.49$, P -value < 0.001 .

Table 4. Summary of predictors for culling reasons identified by multinomial logistic regression (reference = low production)^{1,2}

Outcome ³	Predictor	Level (referent) ⁴	Effect direction ⁵	OR ⁶	95% CI ⁶	P-value
Body condition abnormalities	Breed	MH (H)	↑	2.832	1.389–5.777	0.004
	Lactation stage	Early (late)	↓	0.193	0.092–0.404	<0.001
Feet or leg problems	Lactation stage	Early (late)	↓	0.37	0.179–0.762	0.007
Death	Season	Spring (winter)	↑	2.588	0.946–7.083	0.064
Reproductive problems	Parity group	Second (first)	↓	0.158	0.034–0.745	0.020
		>5 (first)	↓	0.011	0.001–0.364	0.012
	Age (nested, per month)	—	↑	1.059	0.996–1.126	0.066
		Early (late)	↓	0.229	0.096–0.545	0.001
	Lactation stage	Mid (late)	↓	0.207	0.056–0.764	0.018
		>5 (first)	↑	15.705	0.592–416	0.099
Other reasons	Season	Spring (winter)	↑	2.794	0.988–7.904	0.053
	Lactation stage	Mid (late)	↑	2.797	1.108–7.060	0.030

¹Full regression coefficients (B, SE, OR, 95% CI) are reported in the supplemental material, see Notes.

²Holstein, n = 345; Montbéliarde × Holstein, n = 178; Total, n = 523.

³Reference outcome: low production culling in the multinomial model.

⁴Reference levels: H = Holstein, winter, first parities, late lactation.

⁵“↑” = higher odds of this culling reason; “↓” = lower odds.

⁶OR = odds ratio; 95% CI = 95% CI for the OR.

(OR = 2.59; $P = 0.064$). No other predictors reached significance for this outcome in the primary model.

Reproductive problems were strongly associated with parity and lactation stage. Cows in their second parity and in parity ≥ 5 had substantially lower odds of reproductive culling compared with first-parity cows (OR = 0.16 and 0.01, respectively; $P < 0.05$). Early- and mid-lactation cows were less likely to be culled for reproductive problems than late-lactation cows ($P \leq 0.018$). Age nested within parity showed a marginal positive association with reproductive culling risk (OR = 1.06 per month; $P = 0.066$).

For culling classified under other reasons, cows in parity ≥ 5 had higher odds compared with first-parity cows (OR = 15.7; $P = 0.099$), and midlactation was associated

with increased odds of culling relative to late lactation (OR = 2.80; $P = 0.030$). Spring calving showed a near-significant positive association ($P = 0.053$).

Multinomial Logistic Regression Analysis of Culling Reasons (Including BW)

A secondary model excluded the 90 dead cows and additionally included BW to assess the robustness of breed effects and other covariate estimates. The results are presented in Supplemental Table S4, see Notes, and the comparison between the primary and secondary models is shown in Table 5 and illustrated in Supplemental Figure S1, see Notes.

Table 5. Comparison of multinomial logistic regression models with and without BW¹

Outcome category	Predictor	Model excluding BW	Model including BW	Concordance
Body condition abnormalities	Breed (MH vs. H) ²	Significant (↑) ³	Significant (↑)	Consistent
	Lactation stage: Early (late)	Significant (↓)	Significant (↓)	Consistent
Feet or leg problems	Lactation stage: Early (late)	Significant (↓)	Significant (↓)	Consistent
	Breed (MH vs. H)	NS ⁴	Significant (↑)	Strengthened
Reproductive problems	Parity group: Second (first)	Significant (↓)	Significant (↓)	Consistent
	Parity group: >5 (first)	Significant (↓)	Significant (↓)	Consistent
	Lactation stage: Early (late)	Significant (↓)	Significant (↓)	Consistent
	Lactation stage: Mid (late)	Significant (↓)	Significant (↓)	Consistent
	Age (nested, per month)	Near-significant (↑)	Near-significant (↑)	Consistent
	Lactation stage: Mid (late)	Significant (↑)	Significant (↑)	Consistent
Other reasons	Season: Spring (winter)	Near-significant (↑)	Significant (↑)	Strengthened

¹Near-significant indicates $0.05 \leq P < 0.10$. Strengthened indicates increased significance after inclusion of BW. Significant indicates $P < 0.05$. Reference outcome category: low production. Full regression outputs are reported in Supplemental Tables S3 (without BW) and S4 (with BW).

²H = Holstein; MH = Montbéliarde × Holstein.

³“↑” = higher odds of this culling reason; “↓” = lower odds.

⁴NS = not significant.

Overall, the inclusion of BW did not materially alter the direction or significance of most predictors, indicating robust model performance. For body condition abnormalities, breed and lactation stage effects remained significant and consistent across both models. For feet or leg problems, the breed effect became significant after inclusion of BW, suggesting that BW had partially masked the underlying breed-related differences in this outcome. After adjustment for BW, the independent breed effect became apparent. For reproductive problems, parity, lactation stage, and the marginal age effect remained stable, and BW did not substantially modify these associations.

For other culling reasons, the positive association with midlactation remained significant, while the seasonal effect of spring calving strengthened after inclusion of BW. Across all outcomes, no major discrepancies were observed between models, supporting the conclusion that BW acted primarily as a modifying factor rather than a primary driver of breed differences in culling structure.

Cull Price

The factors influencing cull price identified by the linear mixed model are presented in Table 6, and several predictors exhibited significant effects on the dependent variable. To evaluate whether breed differences in cull price were mediated by BW, an additional sensitivity model excluding BW was fitted (Supplemental Table S5, see Notes). The predicted cull prices for the 2 breeds are shown in Supplemental Figure S2, see Notes. Cows culled in 2023 had significantly higher cull prices than those culled in 2024 ($P < 0.001$). BW was positively associated with cull price ($P < 0.001$), indicating that heavier animals tended to have greater economic value at culling. In contrast, breed (H vs. MH), age at culling, and DIM did not show significant effects on the outcome ($P > 0.05$). Importantly, breed remained nonsignificant in the model excluding BW, indicating that the absence

of breed differences in cull price was not attributable to adjustment for BW.

DISCUSSION

The present study compared lactation performance, culling patterns, and economic outcomes between H cows and MH cows under intensive dairy production conditions. Although overall milk yield was largely comparable between the 2 genetic groups, differences were observed in lactation dynamics and, more importantly, in the structure of culling. These findings indicate that similar production outputs can be achieved through distinct biological and management pathways, with important implications for animal survival, risk management, and system sustainability.

Lactation Performance and Production Dynamics

Across lactations, H cows and MH cows exhibited broadly similar total milk yield, confirming that crossbred cows can achieve production levels comparable to pure H cows in high-input systems, a finding consistent with previous reports (Heins et al., 2006b). However, there were some differences in lactation curve characteristics. The MH cows tended to produce less milk at peak, resulting in a flatter and more persistent lactation curve. This pattern aligns with recent findings that Montbéliarde-sired crossbreds exhibit superior lactation persistency compared with pure H cows, compensating for lower peak yields through sustained production in mid-to-late lactation (Houdek et al., 2024).

These different lactation curve characteristics may suggest alternative production strategies rather than inferior performance. A lower peak may reduce metabolic stress in early lactation, which has been associated with improved health and survival in dairy cows (Gross and Bruckmaier, 2019; Caixeta and Omontese, 2021). Therefore, despite similar cumulative yields, the distinct lactation dynamics observed between breeds

Table 6. Parameter estimates of fixed effects in the linear mixed model for cull price (CNY/kg)¹

Parameter	Estimate	SE	df	t-value	P-value	95% CI
Intercept	11.524	0.847	427	13.61	<0.001	9.859–13.189
Breed (MH vs. H)	−0.380	0.338	427	−1.12	0.262	−1.045–0.285
Cull date (2023 vs. 2024)	4.222	0.315	427	13.42	<0.001	3.603–4.840
Age at culling (mo)	−0.003	0.007	427	−0.44	0.662	−0.018–0.011
DIM	−0.002	0.001	427	−1.65	0.100	−0.004–0.000
BW (kg)	0.006	0.002	427	4.21	<0.001	0.003–0.009

¹CNY = Chinese yuan; US\$1 = 7.12 CNY; and €1 = 7.72 CNY.

may partially underpin subsequent differences in culling patterns.

Breed Differences in Culling Structure

The most prominent finding of this study was the distinct exit dynamics between breeds. The H cows experienced a substantially higher proportion of culling due to death, whereas MH cows were predominantly removed for body condition-related reasons. This divergence in culling patterns is in line with previous studies on dairy cattle crossbreeding, which have similarly reported differences in exit dynamics between crossbred and Holstein cows. (Dickinson and Touchberry, 1961; Heins et al., 2012a; Pipino et al., 2023). This distinction is critical from a welfare and economic standpoint: death represents an abrupt, nonrecoverable loss, as mortality events result in zero cull value in the studied production system while body condition-related culling is generally planned, detectable, and manageable through intervention (Orpin and Sibley, 2014). From a systems perspective, the higher incidence of mortality in H cows reflects a greater frequency of abrupt, nonrecoverable loss events that contribute disproportionately to their elevated overall culling risk. In contrast, MH cows tended to exit the herd through pathways that were more predictable and subject to management decisions, despite exhibiting a higher incidence of body condition-related removals. Importantly, this higher incidence of condition-related culling in MH cows did not translate into a higher overall removal risk, indicating that breed differences were driven not only by the reasons for culling but also by the balance between planned and unplanned exits. This pattern suggests, rather than confirms, differences in functional robustness as expressed through exit dynamics under the studied production conditions.

Effects of Season, Parity, and Age on Culling Risk

Seasonal effects were evident for several culling categories, including death, with a tendency toward higher risks observed outside the winter period (De Rensis and Scaramuzzi, 2003; Pinedo et al., 2010; Portner and Heins, 2024). Seasonal associations were observed across culling categories and breeds. Because season was defined based on calving rather than culling, these patterns likely reflect differences in early-lactation conditions and management challenges associated with the calving season, whose consequences may persist and influence culling decisions later in the production cycle. The consistency of seasonal effects across breeds suggests that calving-season conditions act as a general source of variation in culling risk, while potential interactions with genetic background may shape how these effects are ultimately

expressed (Ettema and Santos, 2004; Polsky and von Keyserlingk, 2017).

Both age and parity were associated with culling outcomes, reflecting complementary dimensions of reproductive history and parity-specific aging processes. Although closely related, these variables capture distinct aspects of cow lifecycle progression, with age evaluated within parity strata to reduce redundancy between chronological age and parity. Parity showed biologically meaningful associations with culling risk, consistent with a survivor effect whereby cows that progressed to higher parities represent a more robust subset of the population (Ducrocq, 1994; Pinedo et al., 2010). Within parity groups, parity-specific age was associated with modest changes in the odds of certain culling outcomes, indicating that aging effects were expressed relative to reproductive stage rather than as a uniform chronological process (Adamie et al., 2023). Importantly, parity-specific age was not a significant predictor of mortality in the multinomial framework, suggesting that aging primarily influenced the likelihood and type of nonfatal, management-related culling rather than directly increasing the risk of death. This pattern supports the notion that breed influences the mode, rather than the inevitability, of exit from the herd (Pritchard et al., 2013; Portner and Heins, 2024).

Role of BW in Modulating Culling Risks

Body weight was associated with specific culling outcomes, acting as a modifier of risk expression rather than a primary driver of breed differences. Consistent with previous studies, higher BW was associated with a reduced risk of lameness-related culling, likely due to enhanced structural robustness and digital cushion thickness (Bicalho et al., 2009). However, increased BW was also linked to a modestly higher likelihood of reproductive-related culling, reflecting well-documented trade-offs between physical maintenance and metabolic efficiency (Oltenacu and Broom, 2010). These associations are biologically plausible and reflect trade-offs between physical capacity and metabolic or reproductive efficiency (Drackley, 1999; Bicalho et al., 2009). Crucially, adjustment for BW did not eliminate or reverse the observed breed differences in culling structure. Breed effects on mortality versus nonfatal culling pathways remained directionally and statistically consistent after accounting for BW. This indicates that while BW contributes to variation in the type of problems leading to culling, it does not account for the fundamental divergence between H and MH cows in terms of mortality versus nonfatal culling. Thus, BW functions as a secondary, modulating factor within the broader genetic and physiological framework.

Economic Implications and System Sustainability

Despite clear differences in culling structure, cull prices did not differ significantly between breeds. This finding was robust across model specifications, including a sensitivity analysis excluding BW, which similarly showed no breed effect on cull price. This finding underscores that the economic consequences of culling are driven less by the terminal value of the cow and more by the circumstances under which culling occurs (Hadley et al., 2006; Pinedo et al., 2010). Death culling represents a loss with minimal recovery of value, whereas planned culling allows for at least partial economic return and better integration into herd replacement strategies.

From a sustainability perspective, reducing involuntary losses and shifting culling toward more manageable categories is increasingly important in intensive dairy systems facing rising costs and heightened welfare expectations (Hadley et al., 2006; Pinedo et al., 2010). The lower incidence of death culling observed in MH cows suggests potential advantages in terms of longevity, predictability, and risk management, even when economic returns at culling are comparable between breeds and largely influenced by BW rather than genetic background.

Study Limitations

Several limitations of this study should be acknowledged. First, the data were derived from a single intensive dairy production system, which may limit generalizability to herds operating under different management, nutritional, or environmental conditions. However, this also reduced between-herd variability and allowed breed differences to be evaluated under uniform management. Second, culling reasons were based on farm records and reflected the primary reason assigned at removal; although some degree of misclassification among nonfatal culling categories cannot be excluded, death represents a definitive outcome and constituted a key component of the observed breed differences.

An additional limitation relates to the assumption that identical culling criteria were applied to both breed groups. Although management protocols were consistent at the farm level, the decision to introduce MH crossbreds may have been accompanied by implicit or unintended differences in how culling thresholds were applied to H and MH cows. Such differences are unlikely to affect death as a culling outcome, but they may influence discretionary culling categories, particularly those related to body condition or management judgment.

In addition, the culling category of “body condition abnormalities” used in this study is uncommon in many international recording systems, where standardized body condition scores or more specific disease diagnoses

are typically applied. Therefore, caution is warranted when making direct comparisons of this culling category across countries or production systems.

Finally, nonsignificant associations should be interpreted with caution given the limited sample size, and the observational nature of the study precludes causal inference. The economic assessment focused on cull value rather than comprehensive lifetime profitability, which should be addressed in future research.

CONCLUSIONS

This study demonstrates that similar milk production levels can be achieved by H and MH cows through different biological trajectories, resulting in distinct culling structures. Breed differences were expressed primarily in the mode of exit from the herd rather than in cumulative productivity. These findings highlight the importance of evaluating dairy cattle performance beyond milk yield alone and emphasize culling structure as a complementary indicator of resilience, welfare, and economic stability in modern dairy production systems.

NOTES

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Nonstandard abbreviations used: CNY = Chinese yuan; H = Holstein; MH = Montbéliarde × Holstein; NS = not significant; OR = odds ratio; parameter a = initial yield level; parameter b = ascending slope to peak; parameter c = postpeak decline rate.

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